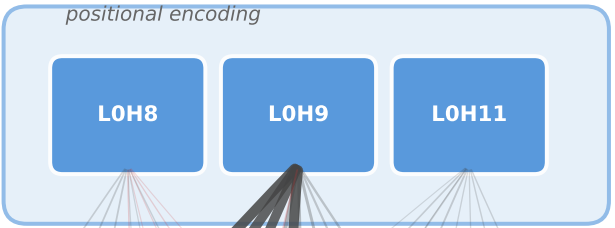


RTI Circuit: 15 Heads in 4 Tiers

Edge width proportional to path-patching effect (302 prompts)

Backbone (L0)

positional encoding



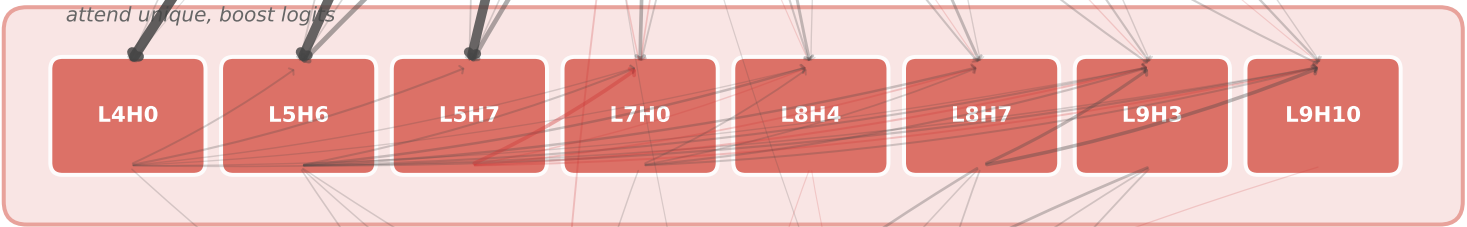
Detector (L4)

previous-token head



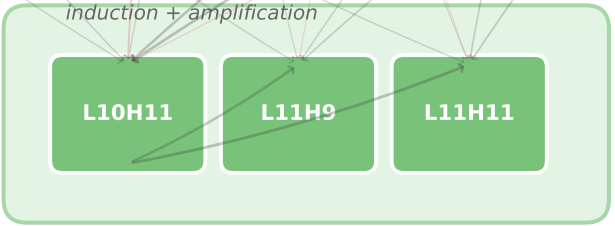
Copier (L4-L9)

attend unique, boost logits



Readout (L10-L11)

induction + amplification



— Positive effect
— Inhibitory effect

Backbone→Detector: 98% recovery
Copier→Readout: 47% recovery
63% of total effect from interactions
L0H9: dominant hub (top 4 edges)